Experiential Learning Project Assessment-I on

"Student Attendance Tracker"

under the course
Python Programming Language (N-PCCCM304P)

Presented By
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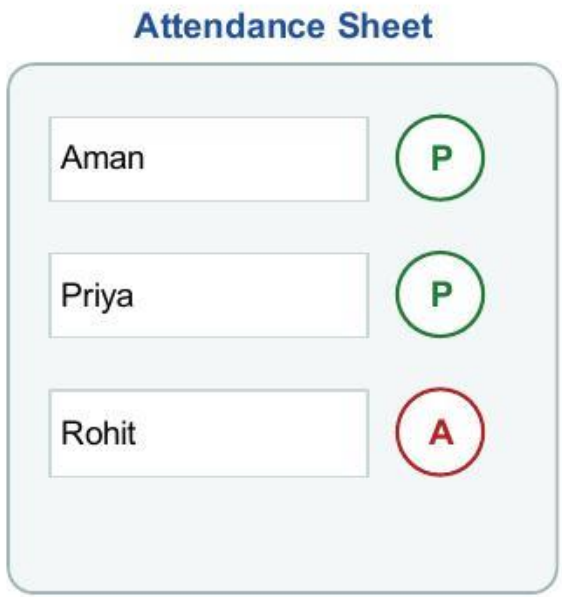
Under the Guidance
of
Ms.Suchita Meskar

Session: 2026-27 (ODD)
Department of CSE (AI&ML)
S. B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT AND RESEARCH, NAGPUR
An Autonomous Institute, Affiliated to RTMNU, Nagpur

Problem Statement :

Student Attendance Tracker

Develop a Python-based attendance management system that records daily student attendance, calculates attendance percentages, identifies defaulters, and generates attendance reports — without file handling or a database.



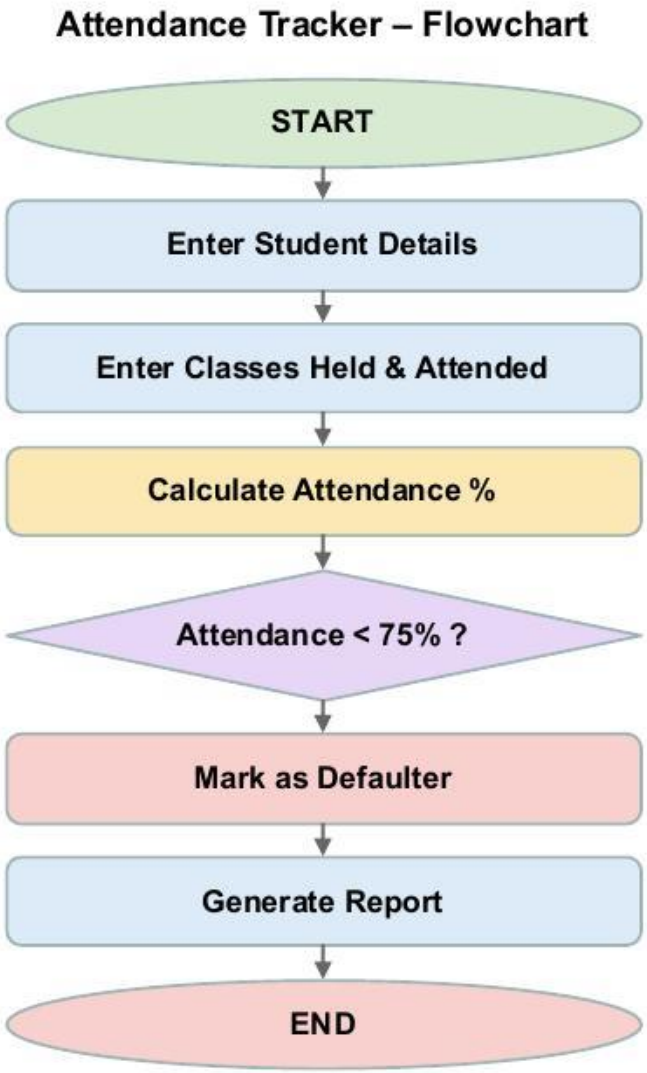
Objectives :

- To develop a Python-based student attendance tracking system.
- To record daily attendance for each student.
- To calculate the attendance percentage for each student.
- To identify students with low attendance (defaulters).
- To generate a clear, formatted attendance report.
- To manage attendance data using only variables and data structures — no files or database.
- To make attendance tracking faster, easier, and more accurate.

Algorithm :

1. Start the program.
2. Enter student details (Name, Roll No.).
3. Enter the total number of classes conducted.
4. Enter the number of classes attended by each student.
5. Calculate the attendance percentage for each student.
6. Compare the percentage with the defaulter threshold (e.g., 75%).
7. Identify students who are defaulters.
8. Generate the attendance report.
9. Display the final attendance report.
10. Stop the program.

Flowchart / Pseudocode :



Pseudocode

```
START
INPUT student list (Name, Roll No.)
INPUT total classes conducted
FOR each student
  INPUT classes attended
  CALCULATE attendance %
  IF % < 75
    MARK as Defaulter
  ELSE
    MARK as Regular
  END IF
END FOR
GENERATE attendance report
DISPLAY report (names, %,
        defaulter status)
END
```

Python Logic Used :

Variables – To store student details, class counts, attendance, and calculated values.

Input/Output – To take attendance data and display the report.

Conditional Statements – Used to identify defaulters based on the attendance threshold.

Loops – Used to process attendance for multiple students.

Lists/Dictionaries – Used to store and organize student-wise attendance data.

Functions – Used to divide the program into reusable modules.

Arithmetic Operations – Used for attendance percentage calculations.

Report Generation – Used to display a formatted attendance report — no file handling or database.

GitHub :

GitHub Account : tanmay386

Repository : student-attendance-tracker

Repository URL : https://github.com/tanmayniwate-386/student-attendance-tracker



Thank You

S. B. Jain Institute of Technology Management and Research

PHASE 2 · PROJECT 70% COMPLETE

Progress Status

70%

of the project completed



Phase 2 review

Completed

- Student add / delete
- Daily attendance marking
- Attendance % calculation
- Defaulter identification
- Basic report

Pending

- Dashboard summary
- Report screen polish
- Input validation
- Final testing

PHASE 2 · PROJECT 70% COMPLETE

Modules Implemented So Far

1 Student Management

Add a student by name and remove a student; each student gets a unique ID.

2 Mark Attendance

Select a date and set each student as Present, Absent or Late.

3 Calculation Logic

Counts present, absent and late days and computes the attendance percentage.

4 Defaulter Check

Compares the percentage with the 75% threshold to flag defaulters.

PHASE 2 · PROJECT 70% COMPLETE

Remaining Work & Next Steps

Dashboard	Show total students, present / absent today and defaulter count.
Report screen	Present, absent, late, total days, attendance % and status for every student.
Validation & testing	Handle empty names and wrong input, then test with multiple days of data.
Final output	Capture screenshots and prepare the final presentation.

PHASE 3 · PROJECT 100% COMPLETE

Project Completed

100%

All modules complete and tested

Dashboard
Total students, present / absent today, defaulters

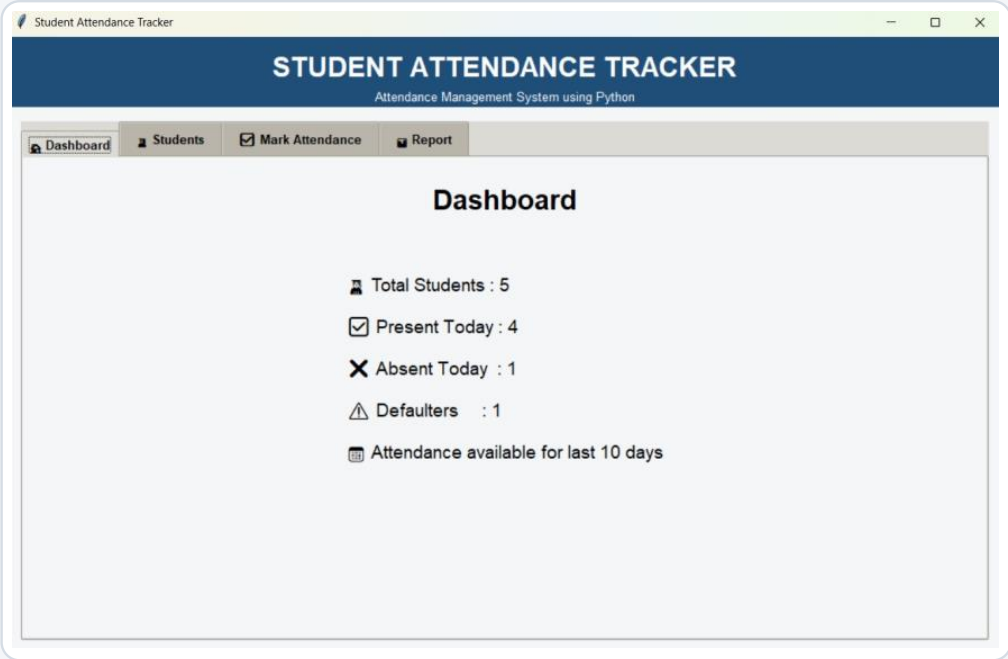
Students
Add and delete students with unique IDs

Mark Attendance
Pick a date; Present → Absent → Late on double-click

Report
Per-student totals, attendance % and status

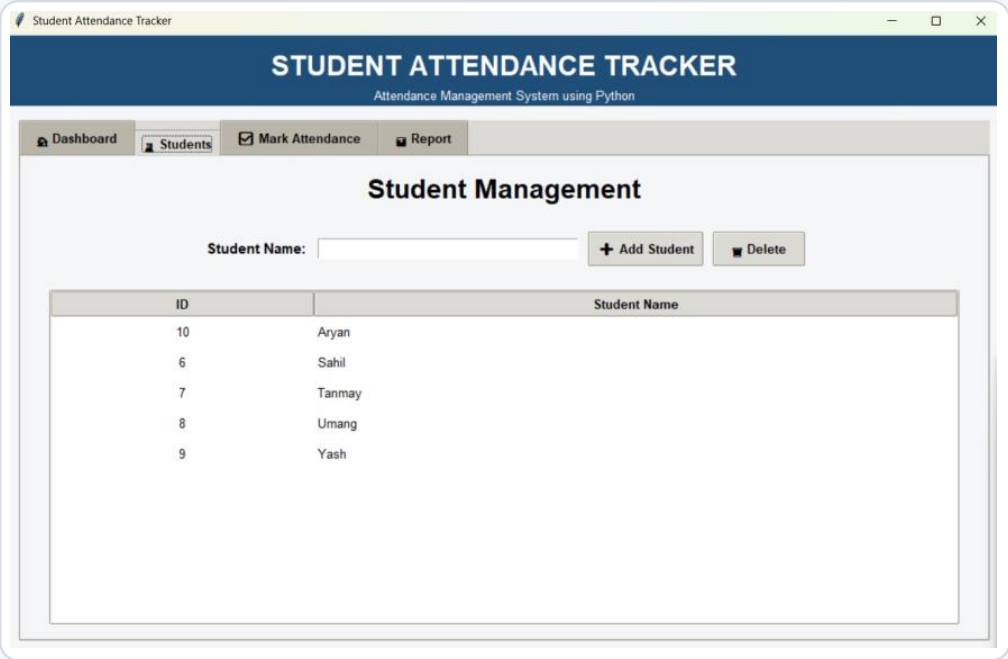
PHASE 3 · PROJECT 100% COMPLETE

Final Output – Dashboard & Students



Dashboard

Live summary of students, today's attendance and defaulters

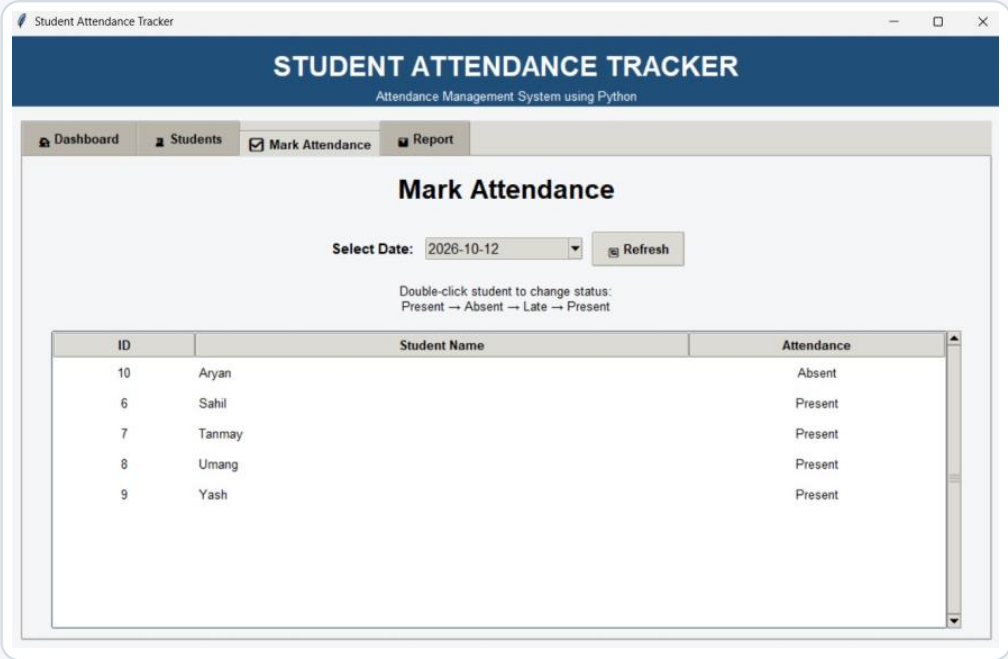


Student Management

Add or delete students from one screen

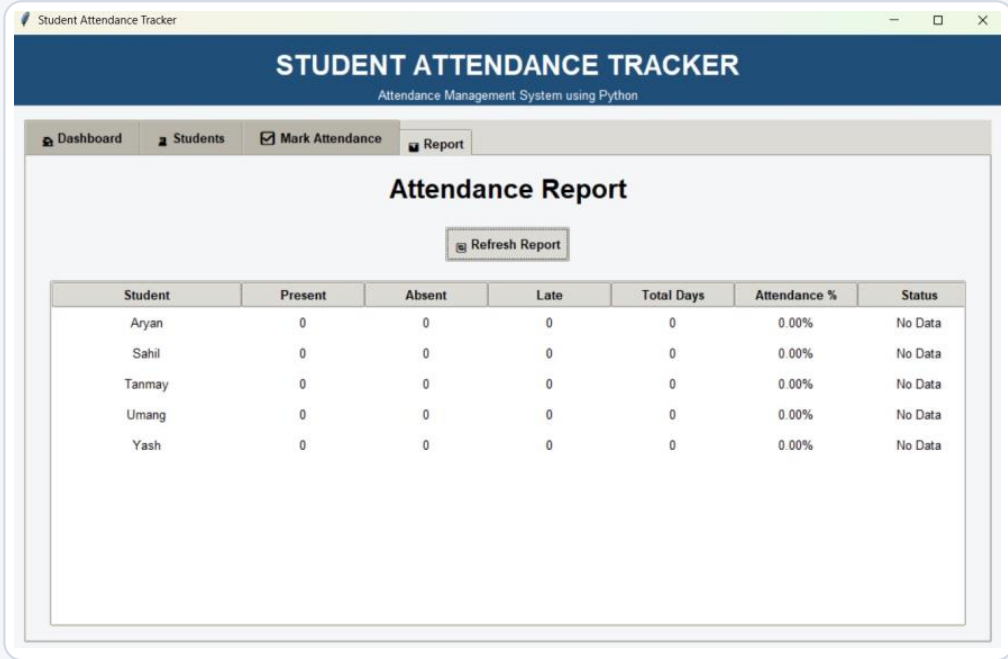
PHASE 3 · PROJECT 100% COMPLETE

Final Output – Attendance & Report



Mark Attendance

Select a date and double-click a student to change status



Attendance Report

Present, absent, late, total days, % and status per student